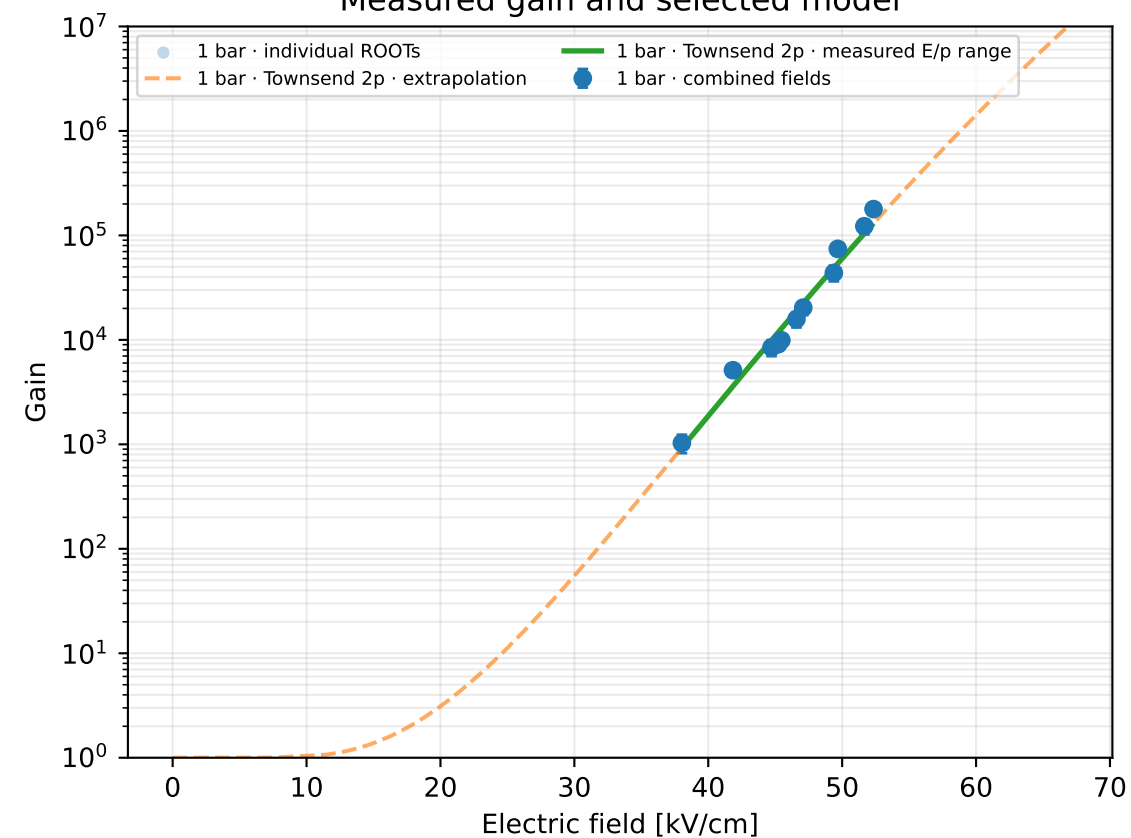
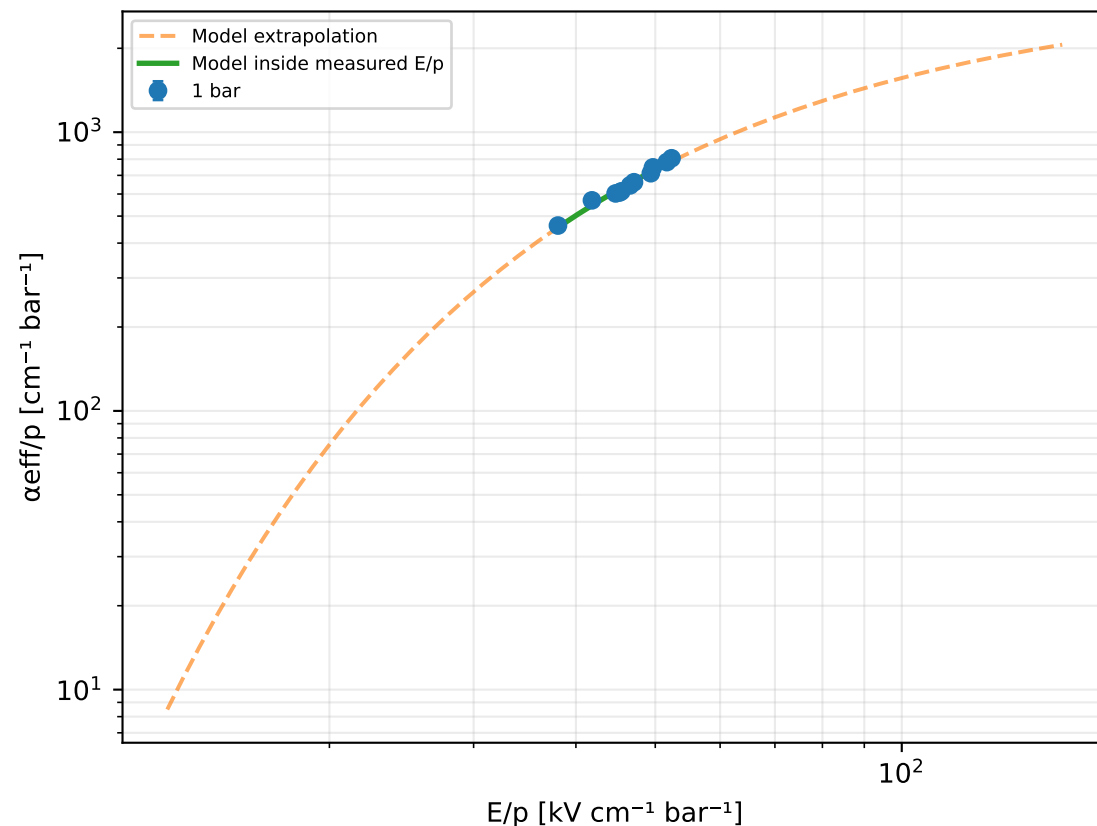


ArlsoCF4 · Ar 95 % / Iso 3 % / CF₄ 2 % · gap 0.15 mm

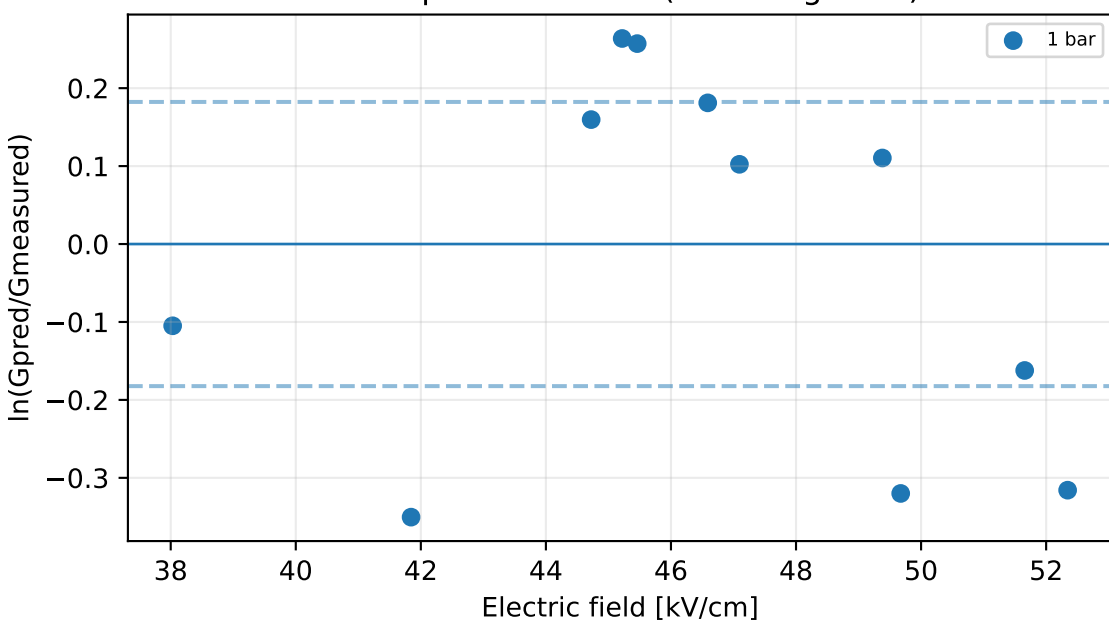
Measured gain and selected model



Reduced Townsend fit

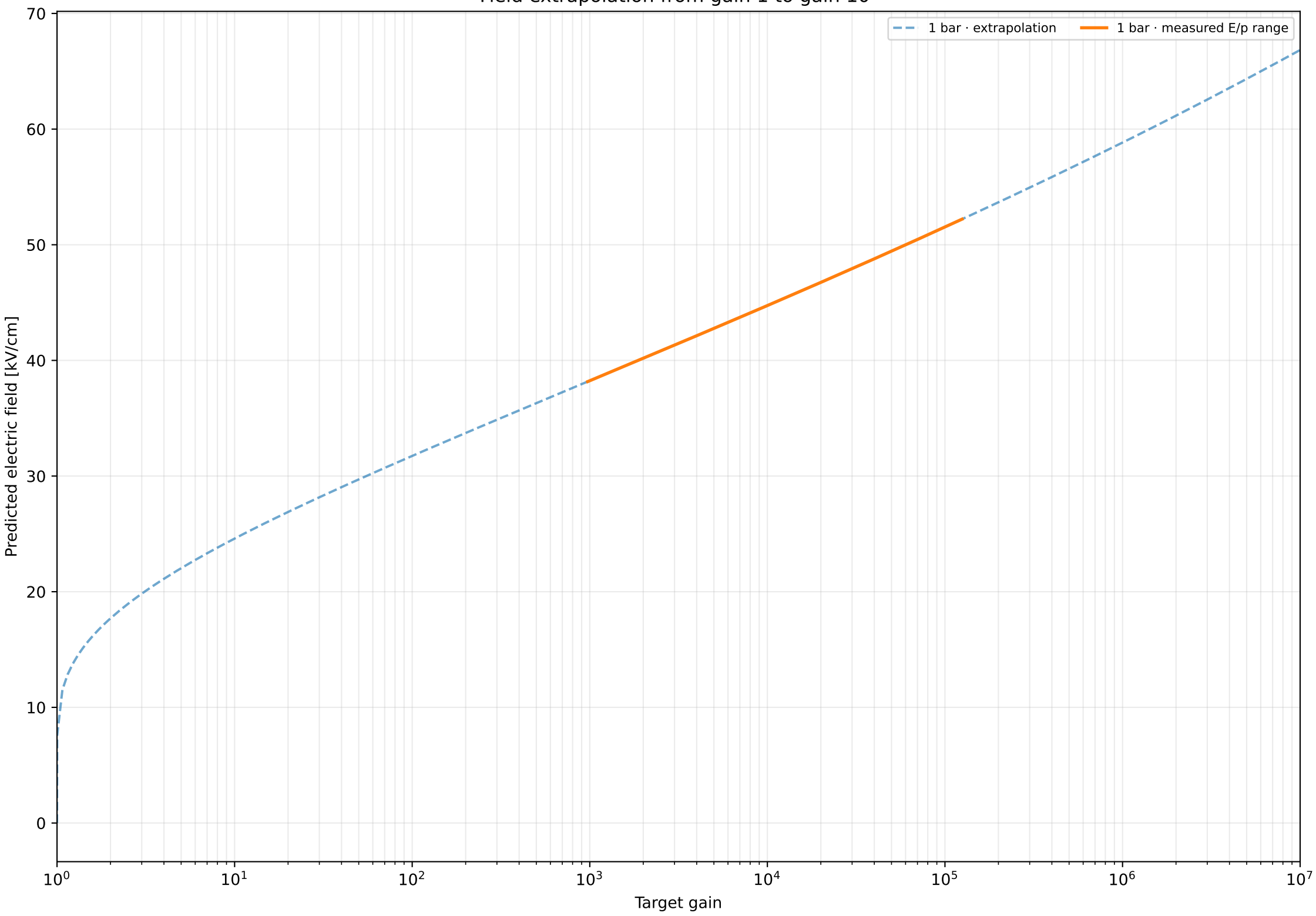


Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 11
Unique physical points: 11
Pressures: 1 bar
Selected model: Townsend 2p
Status: accepted
Training RMSE: $\times 1.024$
Cross-validation median: $\times 1.339$
Cross-validation worst: $\times 1.571$
Reduced-field range: 38.03–52.34 kV cm⁻¹ bar⁻¹
A=3336.9, B=75.737, m=0, n=1
Covariance condition: 562

Field extrapolation from gain 1 to gain 10^7



p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
1	38.031	38.031	1029.55	207	462.458	20	1
1	41.8443	41.8443	5125.15	693	569.461	20	1
1	44.724	44.724	8470.65	1.49e+03	602.958	20	1
1	45.22	45.22	9071.05	992	607.523	20	1
1	45.4602	45.4602	9924.5	1.44e+03	613.517	20	1
1	46.5881	46.5881	15806.15	2.67e+03	644.544	20	1
1	47.095	47.095	20350.9	3.12e+03	661.392	20	1
1	49.3808	49.3808	43797.75	7.52e+03	712.489	20	1
1	49.6741	49.6741	74310.85	9.97e+03	747.734	20	1
1	51.6569	51.6569	122384.4	2.03e+04	780.995	20	1
1	52.3436	52.3436	178597.5	1.91e+04	806.193	20	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-76.9	×1.34	×1.57	yes	—
Townsend 3p	3	-76.7	×1.15	×1.61	no	parameter at bound
Townsend 4p	4	-71.5	—	—	no	ill conditioned covariance, poor cross validation, large cross validation outlier